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INTRODUCTION

PYTHON

Python is a popular programming language. It was created by Guido Van Rossum and released in 1991. It is used for web development, software development, mathematics and system scripting.

Python is used on a server to create web applications along with software to create workflows. It can connect to database systems. Python can be used to handle big data, perform complex mathematics, read and modify files. It is a must learn program for all software engineers because:

- Python is Interpreted – It is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is like PERL and PHP.
- Python is Interactive – You can sit at a Python prompt and interact with the interpreter directly to write your programs.
- Python is Object-Oriented – It supports Object-Oriented style or technique of programming that encapsulates code within objects.
- Python is a Beginner's Language – It is a great language for beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.
- A broad standard library – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Macintosh.
- GUI Programming – It supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.

STRUCTURED QUERY LANGUAGE

(SQL)

Structured query language (SQL) is a programming language for storing and processing information in a relational database. A relational database stores information in a tabular form, with rows and columns representing different data attributes and the different relationships between the data values. SQL can be used to store, update, remove, search, and retrieve data from the database as well as maintain and optimize database performance.

SQL is a popular query language that is used in all types of applications. Data analysts and developers learn and use SQL because it integrates well with different programming languages. SQL was invented in the 1970s based on the relational data model.

It was initially known as the structured English query language (SEQUEL). The term was later shortened to SQL. Oracle, formerly called Relational Software, became the first vendor to offer a commercial SQL relational database management system.

Python SQL Server Integration requires you to build a connection between Python and the SQL server using the pyodbc.connect function and pass a connection string. Pyodbc is a Python open-source module that simplifies access to ODBC (Open Database Connectivity) databases such as Oracle, MySQL, PostgreSQL, SQL Server.

ABOUT THE PROJECT

In today's digital era, computers have become an indispensable part of the banking sector, driving unprecedented advancements. The integration of computer technology has not only streamlined traditional banking processes but also opened new possibilities for banks and their customers. By leveraging the power of computers, the banking industry has experienced remarkable improvements in efficiency, security, customer services, and financial innovation.

Computerization in Banking has led to Faster Transaction Processing, Accurate Financial Calculations, Online Banking Services, 24/7 Access to Accounts, Convenient Payment Options Data Encryption and Protection, Fraud Detection and Prevention, Secure Transactions, Efficient Record Keeping, Data Analytics for Decision Making, Online Loan Applications, Mobile Banking and Digital Wallets and much more.

Having gained an insight into digitalization, we have made Our Computer Project on Banking Management System where in an application has been made using program of Python and SQL to execute simple transactions in Our Bank named The Regal Bank of India.

To start with, we have used the Python and SQL to formulate basic transactions like opening of account, withdrawals, deposits, cards, loans which are enumerated as follows:

First, when the user enters the program, he will have to sign in or create an account. If the user does not have his account in the bank database, then he can create an account by entering his/her name and a password and account number is generated for the user using the randint() function by the program. Once an account is created, to prevent any fake sign-in, the program will exit as soon as the user enters two incorrect passwords. The data of the existing users is stored in the table of the database saved in SQL.

After the user has logged in, the home page appears with a series of functions that it can perform for the user such as Withdrawals, Deposits, Cards, Loans and so on and so forth.

Now, let us discuss how the program is executed for performing each of the above functions.

The first function is to withdraw money from the account. In order to do this, the user must enter his email id to receive the OTP. Once the OTP is received, the user must enter the same before continuing. If the OTP is entered incorrectly twice, the program will exit. After this, the user enters the amount to be withdrawn. Once the amount is withdrawn, that amount is subtracted from the current balance in the account of the user. It is to be noted that the transaction history is updated every time a transaction occurs, and the entire list of transactions is maintained for every user. Here, the smtplib email package has been used, for sending emails to our users for OTPs.

Like any ideal bank, the software allows the user to deposit his amount either as a normal deposit, fixed deposit or recurring deposit. To do this, the user has to use his email and enter the OTP sent by the bank.

In normal deposit, the user enters the amount he would like to deposit, and the changes are made to the account balance.

In a Fixed Deposit, the bank locks the sum of money or the deposit amount for the deposit period providing a certain interest rate for that amount. In our program, interest is set at 12% and the amount is fixed for one year.

Recurring Deposit is the type of deposit that helps people with regular incomes deposit a fixed amount every month into their recurring deposit account and earn interest at the rate applicable to fixed deposits.

After each type of deposit, the program saves the transaction in the history of the deposit.

The third and fourth function allows the user to apply for Cards and Loans. In the cards function, the user selects what type of card he would like to apply for, like debit card, credit card etc. The user then chooses to either apply for a new card or activate an old one. If the user chooses to apply for a new card, the documents needed will be provided on the website and he will have to visit a branch for further details. The activating old card option asks the user to enter his 16-digit card number for activation.

To apply for a loan the user must choose between the loans the bank offers such as Education loan, Home Loan, etc. He will then have to enter a valid phone

number to display the documents required for each type of loan. The program asks the user if he would like to view the forms for the loan chosen. In the affirmative, the user is asked to enter his email id. An email with a URL containing the form is then sent to the user for further action.

The fifth function allows the user to check his current balance in his bank account. The current balance keeps updating based on the money withdrawn and deposited by the user.

The sixth function allows the user to check his transaction history, which is a list that is updated every time the user makes a withdrawal or deposit into his account.

The seventh function is the deposit history which is updated after every fixed deposit or recurring deposit is made or closed by the user.

The eighth function gives the user the Bank's contact information along with the email id.

The ninth function is the logout option. This option first asks the user if he is sure he wants to log out. If yes, the changed balance of the user's bank account is updated in the database and the program is exited. Else, the program continues.

Now let us discuss some of the uses of a Computerized Banking Management System:

1. Automated Transactions: Banks use automated systems to process customer transactions quickly and accurately, thereby reducing the time it takes for customers to transfer money or pay bills.

2. Security: Bank management systems are designed with security measures in place to protect customers' information from unauthorized access or theft. This

includes encryption of data, multi-factor authentication processes, and regular monitoring of system activity.

3. Customer Service: Banking management systems provide helpful features like online banking portals that allow customers to view account balances, and transfer funds between accounts.

4. Data Analytics for Decision Making: Banking management systems play a pivotal role in analyzing data generated by banking operations to extract meaningful information. Through advanced analytics tools and techniques, banks gain a deeper understanding of customer behavior and preferences, thereby making informed business decisions, improving products and services.

SOFTWARE USED

Bank management systems are typically created using programming languages such as Java, C++, and Python. These systems are designed to automate the various processes and functions of a financial institution, such as customer management, accounting, and loan processing.

The systems can be customized to suit the specific needs and requirements of the bank. Additionally, banks often employ third-party software and services to help manage their operations, but as this particular project is not a very software intensive one, we have opted for the very basic IDLE platform (64-bit, Version-3.8) for this project and mysql package (Server version: 5.7.21-log MySQL Community Server)

IDLE (Integrated Development and Learning Environment) is a Python development environment designed specifically for beginners. It is included with the standard Python distribution and is cross-platform, running on Windows, Mac OS X, and Linux. IDLE has a simple and intuitive graphical user interface (GUI) that makes it easy for beginners to get started with programming in Python. It includes a text editor for writing and editing code, a debugger for debugging code, and a shell for running code.

MySQL is a relational database management system that uses SQL. SQL is primarily used to query and operate database systems. MySQL Database is a client/server system that consists of a multithreaded SQL server that supports different back ends, several different client programs and libraries, administrative tools, and a wide range of application-programming interfaces (APIs).

HARDWARE USED

This project does not demand anything in particular. We just need a basic laptop which can run IDLE. Assuming that a laptop or desktop has these basic specifications, we are good to go:

- Processor: Intel Core i3 or AMD Ryzen 3
- RAM: 4GB DDR4
- Storage: 128GB SSD
- Display: 15.6-inch HD (1366x768)
- Graphics: Integrated Intel UHD Graphics 620
- Connectivity: Wi-Fi, Bluetooth, USB 3.1, HDMI
- Battery: 4-cell Lithium-ion

For this project we used a Dell laptop with specifications of:

- Processor: Intel Core i5
- RAM: 4GB DDR4
- Storage: 512GB SSD
- Display: 17.3-inch HD
- Graphics: Integrated Intel UHD Graphics 620
- Connectivity: Wi-Fi, Bluetooth, USB 3.1, HDMI
- Battery: 4-cell Lithium-ion

SOURCE CODE

```
import mysql.connector as m
import random
import os
import math
import smtplib
from email.mime.text import MIMEText

mycon=m.connect(host="localhost", user ="root", password="123456",
database = "BMS")
cur=mycon.cursor()
cur.execute("use BMS")
z=1

# Welcome page

print("Choose one of the following:")
print("1. Create account:")
print("2. Sign in:")
signin_choice= (input("Enter you choice: "))

#Creating the Account and saving the same in the Database

if signin_choice =='1':
```

```
    Name=input("Enter you name: ")
    Acc=random.randint(60000,70000)
    Pass=random.randint(200,999)
    print("Your Account Number is: ",Acc)
    print("Your Password is: ",Pass)
    print("Please enter the amount to be deposited into your account")
    print("(Minimum Amount 10000):")
    Money=float(input())
    q="insert into client(Accountnum, Name, Password, Totalbal) values
({},'{}',{},{})".format(Acc, Name, Pass, Money)
    cur.execute(q)
    mycon.commit()
    print("Welcome to the REGAL BANK OF INDIA ")
    CurrBal=Money

#Signing in

elif signin_choice=='2':
    x=input("Enter your Account Number: ")
    query="select *  from client where accountnum = %s" %(x,)
    cur.execute(query)
    data=cur.fetchall()
    if len(data) == 0:
        print("Invalid Account Number. ")
        exit()
    else:
        for i in range(1,3):
            if i == 1:
```

```
        for a in data:
            print("Enter your password:")
            Pass=int(input())
            if a[2] == Pass:
                print("Welcome to the REGAL BANK OF INDIA ")
                z=z+2
                CurrBal=a[3]
                break

            else:
                z=z+1
                print("Invalid Password. Please try again.")
                print("(This is your last chance!)")
    elif i==2 and z<3:
        print("Enter your password:")
        Pass=int(input())
        for a in data:
            if a[2]==Pass:
                print("Welcome!!!")
                CurrBal=a[3]
                break
            else:
                print("Invalid password.")
                print("(You have exceeded your number of attempts.)")
                exit()

#Wrong option selected
```

```
else:
    print("Invalid Choice.")
    exit()


#_____________________________________


#Home Page
Transactions = []
Savings = []

while True:
    print("")
    print("Choose one of the following:")
    print("1. Withdraw")
    print("2. Deposit")
    print("3. Cards")
    print("4. Loans")
    print("5. Check current balance")
    print("6. Transaction History")
    print("7. Deposits History")
    print("8. Contact Us")
    print("9. Logout")
    m=1
    n=1
    DepT=0
    WithD=0
```

```
Min=1000.0

choice= int(input("Enter your choice: "))
#Withdraw
if choice==1:
    digits="0123456789"
    OTP=""
    for i in range(6):
        OTP+=digits[math.floor(random.random()*10)]
    otp = OTP + " is your OTP"
    msg= otp
    s = smtplib.SMTP('smtp.gmail.com', 587)
    s.starttls()
    s.login("regalbankofindia@gmail.com", "jxpfgfgwqbmfpmds")
    emailid = input("Enter your email: ")
    s.sendmail('&&&&&&&&&&&&',emailid,msg)
    a = input("Enter Your OTP: ")
    if a == OTP:
        print("Verified")

    else:
        print("Please Check your OTP again")
        a = input("Enter Your OTP: ")
        if a == OTP:
            print("Verified")
        else:
            print("Access Denied")
            exit()
```

```
    print("Current Balance in INR: ",CurrBal)
    print("Minimum Balance in INR: ",Min)
    WithD=float(input("Enter amount to be withdrawn: "))
    CurrBal=CurrBal-WithD
    print("Current Balance in INR: ",CurrBal)
    Transactions.append(("Withdrawn",WithD))

#Deposit
elif choice==2:
    digits="0123456789"
    OTP=""
    for i in range(6):
        OTP+=digits[math.floor(random.random()*10)]
    otp = OTP + " is your OTP"
    msg= otp
    s = smtplib.SMTP('smtp.gmail.com', 587)
    s.starttls()
    s.login("regalbankofindia@gmail.com", "jxpfgfgwqbmfpmds")
    emailid = input("Enter your email: ")
    s.sendmail('&&&&&&&&&&&&',emailid,msg)
    a = input("Enter Your OTP: ")
    if a == OTP:
        print("Verified")

    else:
        print("Please Check your OTP again")
        a = input("Enter Your OTP: ")
```

```
    if a == OTP:
        print("Verified")
    else:
        print("Access Denied")
        exit()
print("Choose one of the following:")
print("1.Normal Deposit")
print("2.Fixed Deposit")
print("3.Recurring Deposit")
choice3= int(input("Enter your choice: "))
if choice3==1:
    print("Current Balance in INR: ",CurrBal)
    print("Minimum Balance in INR: ",Min)
    DepT=float(input("Enter amount to be deposited: "))
    CurrBal=CurrBal+DepT
    print("Current Balance in INR: ",CurrBal)
    Transactions.append(("Deposited",DepT))
elif choice3==2:
    print("The rate of Interest per annum: 12%")
    print("Current Balance: ",CurrBal)
    print("Minimum Balance: ",Min)
    DepT=float(input("Enter amount to be deposited: "))
    Annual_calctotal=DepT+(DepT*1*12/100)
    print("Expected Total Balance-Annually:",Annual_calctotal)
    Savings.append(("Savings - Fixed Deposit",DepT))
elif choice3==3:
    print("The rate of Interest per annum: 12%")
```

```
        print("Current Balance: ",CurrBal)
        print("Minimum Balance: ",Min)
        DepT=float(input("Enter amount to be deposited: "))
        print("Choose one of the following")
        print("1.Monthly")
        print("2.Quaterly")
        print("3.Annually")
        choice2= int(input("Enter your choice: "))
        if choice2==1:
            Month_calctotal=(DepT*((1+12/(m*12))**m*1))
            print("Expected Total Balance-Monthly ",Month_calctotal)
        elif choice2==2:
            Quarter_calctotal=(DepT*((1+12/(m*4))**m*1))
            print("Expected Total Balance-Quaterly",Quarter_calctotal)
        elif choice2==3:
            Annual_calctotal=(DepT*((1+12/(m*1))**m*1))
            print("Expected Total Balance-Annually",Annual_calctotal)
        else:
            print("Invalid choice.")
        Savings.append(("Savings - Recurring Deposit",DepT))
        m=m+1
    else:
        print("Invalid Choice. Try again.....")

#Cards
elif choice==3:
    print("Choose one of the following:")
```

```
print("1.Debit Card")
print("2.Credit Card")
print("3.Business Card")
choice3= int(input("Enter your choice: "))
if choice3==1:
    print("Choose one of the following:")
    print("1.Activating old card")
    print("2.Applying for New Card")
    choice4= int(input("Enter your choice:"))
    if choice4==1:
        digits="0123456789"
        OTP=""
        for i in range(6):
            OTP+=digits[math.floor(random.random()*10)]
        otp = OTP + " is your OTP"
        msg= otp
        s = smtplib.SMTP('smtp.gmail.com', 587)
        s.starttls()
        s.login("regalbankofindia@gmail.com", "jxpfgfgwqbmfpmds")
        emailid = input("Enter your email: ")
        s.sendmail('&&&&&&&&&&&&',emailid,msg)
        a = input("Enter Your OTP: ")
        if a == OTP:
            print("Verified")
        else:
            print("Please Check your OTP again")
            a = input("Enter Your OTP: ")
```

```
            if a == OTP:
                print("Verified")
            else:
                print("Access Denied")
                exit()
        Card=int(input("Enter your 16 digit card number:"))
        if Card>=1000000000000000 and Card<=9999999999999999:
            print("The card will be activated soon")
        else:
            print("Invalid Card Number...")
    elif choice4==2:
        print("DOCUMENTS REQUIRED")
        print("a)Aadhar Card")
        print("b)PAN Card")
        print("c)Driving License")
        print("d)Social security card")
        print("The Application will be available at home branch")
        print("")
       print("VISIT THE HOME BRANCH FOR FURTHER DETAILS")
elif choice3==2:
    print("Choose one of the following:")
    print("1.Activating old card")
    print("2.Applying for New Card")
    choice4= int(input("Enter your choice:"))
    if choice4==1:
        digits="0123456789"
        OTP=""
```

```
    for i in range(6):
        OTP+=digits[math.floor(random.random()*10)]
    otp = OTP + " is your OTP"
    msg= otp
    s = smtplib.SMTP('smtp.gmail.com', 587)
    s.starttls()
    s.login("regalbankofindia@gmail.com", "jxpfgfgwqbmfpmds")
    emailid = input("Enter your email: ")
    s.sendmail('&&&&&&&&&&&&',emailid,msg)
    a = input("Enter Your OTP: ")
    if a == OTP:
        print("Verified")
    else:
        print("Please Check your OTP again")
        a = input("Enter Your OTP: ")
        if a == OTP:
            print("Verified")
        else:
            print("Access Denied")
            exit()
    Card=int(input("Enter your 16 digit card number:"))
    if Card>=1000000000000000 and Card<=9999999999999999:
        print("The card will be activated soon")
    else:
        print("Invalid Card Number...")
elif choice4==2:
    print("DOCUMENTS REQUIRED")
```

```
        print("a)Aadhar Card")
        print("b)PAN Card")
        print("c)Driving License")
        print("d)Social security card")
        print("The Application will be available at home branch")
        print("")
       print("VISIT THE HOME BRANCH FOR FURTHER DETAILS")
elif choice3==3:
    print("Choose one of the following:")
    print("1.Activating old card")
    print("2.Applying for New Card")
    choice4= int(input("Enter your choice:"))
    if choice4==1:
        digits="0123456789"
        OTP=""
        for i in range(6):
            OTP+=digits[math.floor(random.random()*10)]
        otp = OTP + " is your OTP"
        msg= otp
        s = smtplib.SMTP('smtp.gmail.com', 587)
        s.starttls()
        s.login("regalbankofindia@gmail.com", "jxpfgfgwqbmfpmds")
        emailid = input("Enter your email: ")
        s.sendmail('&&&&&&&&&&&&',emailid,msg)
        a = input("Enter Your OTP: ")
        if a == OTP:
            print("Verified")
```

```
            else:
                print("Please Check your OTP again")
                a = input("Enter Your OTP: ")
                if a == OTP:
                    print("Verified")
                else:
                    print("Access Denied")
                    exit()
            Card=int(input("Enter your 16 digit card number:"))
            if Card>=1000000000000000 and Card<=9999999999999999:
                print("The card will be activated soon")
            else:
                print("Invalid Card Number...")
        elif choice4==2:
            print("DOCUMENTS REQUIRED")
            print("a)Aadhar Card")
            print("b)PAN Card")
            print("c)Driving License")
            print("d)Social security card")
            print("The Application will be available at home branch")
            print("")
           print("VISIT THE HOME BRANCH FOR FURTHER DETAILS")
    else:
        print("Invalid Choice. Try again.....")

#Loans
elif choice==4:
```

```
print("Choose one of the following:")
print("1.Education Loans")
print("2.Home Loans")
print("3.Auto Loans")
choice3= int(input("Enter your choice: "))
if choice3==1:
    ch= int(input("Enter your phone number: "))
    if ch>=1000000000 and ch<=9999999999:
        print("")
        print("DOCUMENTS REQUIRED")
        print("a)Application form with photograph")
        print("b)Identity and Residence proof")
        print("c)Last 6 months bank statements")
        print("d)Processing fee cheque")
        print("e)Educational Qualifications Certificate")
        print("")
        print("VISIT YOUR HOME BRANCH FOR FURTHER DETAILS")
        r=input("Would you like us to send you the forms (Y/N)")
        if r=="Y":
            subject = "Education Loan Form"
            body = "URL - https://sites.google.com/view/regalbankofindia-educationloan/home"
            sender = "RegalBankofIndia@gmail.com"
            recipients = input("Enter your email:")
            password = "jxpfgfgwqbmfpmds"
            def send_email(subject, body, sender, recipients, password):
                msg = MIMEText(body)
                msg['Subject'] = subject
```

```
                msg['From'] = sender
                msg['To'] = ', '.join(recipients)
               with smtplib.SMTP_SSL('smtp.gmail.com', 465) as smtp_server:
                    smtp_server.login(sender, password)
                    smtp_server.sendmail(sender, recipients, msg.as_string())
                print("Message sent!")
            send_email(subject, body, sender, recipients, password)
        else:
            print("Invalid phone number...")

    if choice3==2:
        ch= int(input("Enter your phone number: "))
        if ch>=1000000000 and ch<=9999999999:
            print("")
            print("DOCUMENTS REQUIRED")
            print("a)Application form with photograph")
            print("b)Identity and Residence proof")
            print("c)Last 6 months bank statements")
            print("d)Processing fee cheque")
            print("e)Educational Qualifications Certificate")
            print("")
           print("VISIT YOUR HOME BRANCH FOR FURTHER DETAILS")
            r=input("Would you like us to send you the forms (Y/N)")
            if r=="Y":
                subject = "Home Loan Form"
                body = "URL - https://sites.google.com/view/regalbankofindia-
homeloan/home"
                sender = "RegalBankofIndia@gmail.com"
```

```
            recipients = input("Enter your email:")
            password = "jxpfgfgwqbmfpmds"
            def send_email(subject, body, sender, recipients, password):
                msg = MIMEText(body)
                msg['Subject'] = subject
                msg['From'] = sender
                msg['To'] = ', '.join(recipients)
              with smtplib.SMTP_SSL('smtp.gmail.com', 465) as smtp_server:
                    smtp_server.login(sender, password)
                    smtp_server.sendmail(sender, recipients, msg.as_string())
                print("Message sent!")
            send_email(subject, body, sender, recipients, password)
    else:
        print("Invalid phone number...")

if choice3==3:
    ch= int(input("Enter your phone number: "))
    if ch>=1000000000 and ch<=9999999999:
        print("")
        print("DOCUMENTS REQUIRED")
        print("a)Application form with photograph")
        print("b)Identity and Residence proof")
        print("c)Last 6 months bank statements")
        print("d)Processing fee cheque")
        print("e)Educational Qualifications Certificate")
        print("")
       print("VISIT YOUR HOME BRANCH FOR FURTHER DETAILS")
```

```
            r=input("Would you like us to send you the forms (Y/N)")
            if r=="Y":
                subject = "Auto Loan Form"
                body = "URL - https://sites.google.com/view/regalbankofindia-
autoloan/home"
                sender = "RegalBankofIndia@gmail.com"
                recipients = input("Enter your email:")
                password = "jxpfgfgwqbmfpmds"
                def send_email(subject, body, sender, recipients, password):
                    msg = MIMEText(body)
                    msg['Subject'] = subject
                    msg['From'] = sender
                    msg['To'] = ', '.join(recipients)
                  with smtplib.SMTP_SSL('smtp.gmail.com', 465) as smtp_server:
                        smtp_server.login(sender, password)
                        smtp_server.sendmail(sender, recipients, msg.as_string())
                    print("Message sent!")
                send_email(subject, body, sender, recipients, password)
            else:
                print("Invalid phone number...")

    #Current Balance
    elif choice==5:
        print("Your Current balance is: ",CurrBal)

    #Transaction History
    elif choice==6:
        print("")
```

```
        print("TRANSACTION HISTORY")
        for j in range(0,len(Transactions)):
            print(Transactions[j])

    #Deposit History
    elif choice==7:
        print("")
        print("DEPOSITS HISTORY")
        for k in range(0,len(Savings)):
            print(Savings[k])

    #Contact Us
    elif choice==8:
        print("")
        print("CONTACT US")
        print("Phone Numbers: ")
        print("   9980259715")
        print("   6364336888")
        print("Email: ")
        print("   RegalBankofIndia@gmail.com")

    #Logout
    elif choice==9:
        print("")
        log = input("Are you sure you want to logout? (Y/N)")
        if log=='Y':
            qu="update client set totalbal = %s where Password = %s"%(CurrBal,
Pass)
```

```
        cur.execute(qu)
        mycon.commit()
        exit()

#Invalid Choice
else:
    print("Invalid choice. Try again.....")
```

OUTPUT

```
Choose one of the following:
1. Create account:
2. Sign in:
Enter you choice: 1
Enter you name: Anish Saranath
Your Account Number is:  65375
Your Password is:  326
Please enter the amount to be deposited into your account
(Minimum Amount 10000):
28000
Welcome to the REGAL BANK OF INDIA
```

```
Choose one of the following:
1. Create account:
2. Sign in:
Enter you choice: 2
Enter your Account Number: 71520
Enter your password:
128
Welcome to the REGAL BANK OF INDIA
```

```
Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 1
Enter your email:              @gmail.com
Enter Your OTP: 491365
Verified
Current Balance in INR:  100000000
Minimum Balance in INR:  1000.0
Enter amount to be withdrawn: 1000
Current Balance in INR:  99999000.0
```

```
Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 2
Enter your email:                @gmail.com
Enter Your OTP: 904535
Verified
Choose one of the following:
1.Normal Deposit
2.Fixed Deposit
3.Recurring Deposit
Enter your choice: 1
Current Balance in INR:  99999000.0
Minimum Balance in INR:  1000.0
Enter amount to be deposited: 1000
Current Balance in INR:  100000000.0
```

```
Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 2
Enter your email:                @gmail.com
Enter Your OTP: 676307
Verified
Choose one of the following:
1.Normal Deposit
2.Fixed Deposit
3.Recurring Deposit
Enter your choice: 2
The rate of Interest per annum: 12%
Current Balance:  99999000.0
Minimum Balance:  1000.0
Enter amount to be deposited: 1000
Expected Total Balance-Annually: 1120.0
```

```
Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 2
Enter your email:                @gmail.com
Enter Your OTP: 165798
Verified
Choose one of the following:
1.Normal Deposit
2.Fixed Deposit
3.Recurring Deposit
Enter your choice: 3
The rate of Interest per annum: 12%
Current Balance:  100000000.0
Minimum Balance:  1000.0
Enter amount to be deposited: 1000
Choose one of the following
1.Monthly
2.Quaterly
3.Annually
Enter your choice: 2
Expected Total Balance-Quaterly 4000.0
```

Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 3
Choose one of the following:
1.Debit Card
2.Credit Card
3.Business Card
Enter your choice: 1
Choose one of the following:
1.Activiting old card
2.Applying for New Card
Enter your choice:1
Enter your email: @gmail.com
Enter Your OTP: 204234
Verified
Enter your 16 digit card number:1357246808649753
The card will be activated soon

Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 3
Choose one of the following:
1.Debit Card
2.Credit Card
3.Business Card
Enter your choice: 1
Choose one of the following:
1.Activiting old card
2.Applying for New Card
Enter your choice:2
DOCUMENTS REQUIRED
a)Aadhar Card
b)PAN Card
c)Driving License
d)Social security card
The Application will be available at home branch

VISIT THE HOME BRANCH FOR FURTHER DETAILS

Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 4
Choose one of the following:
1.Education Loans
2.Home Loans
3.Auto Loans
Enter your choice: 1
Enter your phone number: 9876543210

DOCUMENTS REQUIRED
a)Application form with photograph
b)Identity and Residence proof
c)Last 6 months bank statements
d)Processing fee cheque
e)Educational Qualifications Certificate

VISIT YOUR HOME BRANCH FOR FURTHER DETAILS
Would you like us to send you the forms (Y/N)Y
Enter your email: @gmail.com
Message sent!

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regalbankofindia@gmail.com

URL - https://sites.google.com/view/regalbankofindia-educationloan/home

Reply | Reply all | Forward

Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 4
Choose one of the following:
1.Education Loans
2.Home Loans
3.Auto Loans
Enter your choice: 2
Enter your phone number: 9876543210

DOCUMENTS REQUIRED
a)Application form with photograph
b)Identity and Residence proof
c)Last 6 months bank statements
d)Processing fee cheque
e)Educational Qualifications Certificate

VISIT YOUR HOME BRANCH FOR FURTHER DETAILS
Would you like us to send you the forms (Y/N)Y
Enter your email: @gmail.com
Message sent!

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regalbankofindia@gmail.com

URL - https://sites.google.com/view/regalbankofindia-homeloan/home

Reply | Reply all | Forward

Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 4
Choose one of the following:
1.Education Loans
2.Home Loans
3.Auto Loans
Enter your choice: 3
Enter your phone number: 9876543210

DOCUMENTS REQUIRED
a)Application form with photograph
b)Identity and Residence proof
c)Last 6 months bank statements
d)Processing fee cheque
e)Educational Qualifications Certificate

VISIT YOUR HOME BRANCH FOR FURTHER DETAILS
Would you like us to send you the forms (Y/N)Y
Enter your email: @gmail.com
Message sent!

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regalbankofindia@gmail.com

URL - https://sites.google.com/view/regalbankofindia-autoloan/home

Reply | Reply all | Forward

Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 5
Your Current balance is: 19150.0

Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 6

TRANSACTION HISTORY
('Deposited', 10000.0)
('Withdrawn', 850.0)

Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 7

DEPOSITS HISTORY
('Savings - Fixed Deposit', 2100.0)

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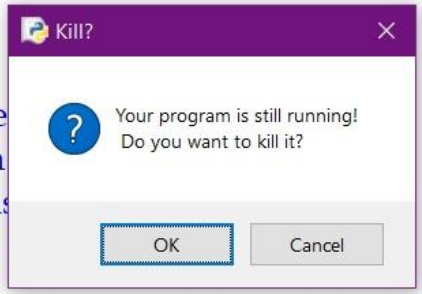
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check current balance
6. Transaction History
7. Deposits History
8. Contact Us
9. Logout
Enter your choice: 8

CONTACT US
Phone Numbers:
  9980259715
  6364336888
Email:
  RegalBankofIndia@gmail.com

Choose one of the following:
1. Withdraw
2. Deposit
3. Cards
4. Loans
5. Check curre
6. Transaction
7. Deposits His
8. Contact Us
9. Logout
Enter your choice: 9



Are you sure you want to logout? (Y/N)Y

CONCLUSION

Thus, Our Python SQL integrated program on Banking has undergone some significant improvements from the previous version, making it even better than before. With new updates and enhancements, the program now offers even more comprehensive functionality and a more intuitive user interface.

Making the database dynamic was an efficient and reliable solution for handling financial transactions and customer information. Its user-friendly interface and robust features make it an ideal choice for our bank. By implementing real-time updates and making it easier for our users to input and access data, we have improved the efficiency and accuracy of our banking system.

To make the user experience more efficient, the OTP option was implemented for all financial transactions. This was a great improvement to the program, making it more secure.

Adding forms in URLs that can easily be downloaded or printed was also a great addition to this program. It made for a more user-friendly and convenient interface, allowing users to access important information at their own pace and in a format that suits them best. This was another way we enhanced the program and enhanced the user's experience.

Finally, we could say that Computers have undoubtedly revolutionized the banking industry, transforming the way banks operate and interact with their customers. Automation, efficiency, enhanced customer experience, improved security measures, data management, and innovative financial services are just some of the benefits brought by computers in banking. As technology continues to advance, the future holds even more possibilities for further transformation and innovation in the banking sector.

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